Circular motion

IGCSE | 0625 Ch.3 | Chapter 3: Forces and Motion | Lesson 3.7 | Syllabus 1.5.1, 6.1.1

Goal: Explain why moving in a circle at steady speed needs a force towards the centre, and how that force depends on speed, mass and radius.

01 Key definitions

Motion in a circle (syllabus) Supplement (Extended) only [syllabus wording]
"Describe, qualitatively, motion in a circular path due to a force perpendicular to the motion as: (a) speed increases if force increases, with mass and radius constant; (b) radius decreases if force increases, with mass and speed constant; (c) an increased mass requires an increased force to keep speed and radius constant."

The syllabus adds: "F = mv²/r is not required."

Uniform circular motion [paraphrase]
Moving round a circle at constant speed. The direction of motion changes all the time, so the velocity changes and the object is accelerating, even though its speed does not change.

Centripetal force [paraphrase]
The resultant force on an object moving in a circle. It acts towards the centre, at 90° to the velocity, so it changes the direction of motion but not the speed. A real force supplies it: tension (ball on a string), friction (car on a bend) or gravity (Moon, satellites).

"Centripetal" = centre-seeking: a job title, not a new force.

02 Core equations

Speed round a circle / average orbital speed Supplement (Extended) only

$$v = \frac{2\pi r}{T}$$

v	speed (average orbital speed)	$\mathrm{m\,s^{-1}}$
r	radius of the circle or orbit	m
T	period: time for one complete circle	s

Syllabus 6.1.1: recall and use. One lap (2πr) ÷ time.

Centripetal force Not examined in 0625 · A Level Ch.16

$$F = \frac{mv^2}{r}$$

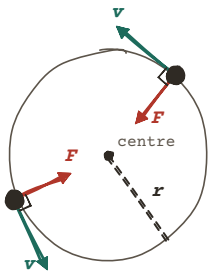
F	resultant (centripetal) force	N
m	mass	kg
v	speed	$\mathrm{m\,s^{-1}}$
r	radius	m

Shows why (a)-(c) hold: more v or m, or less r, needs more F.

03 Diagrams

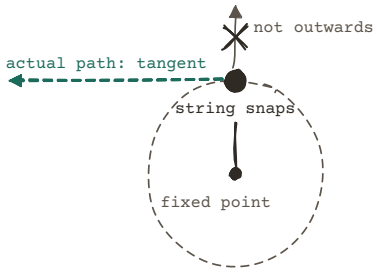
force — velocity

Fig. 1 · Velocity and force



Velocity is along the tangent; the resultant force points to the centre, always at 90° to it. Same-length arrows: steady speed, changing direction.

Fig. 2 · If the string breaks



No inward force, so nothing turns the ball: it moves off in a straight line along the tangent (Newton's first law), not outwards.

04 Worked example

Orbital period from speed and radius Supplement (Extended) only

A satellite orbits in a circle of radius 7.0 × 10⁶ m at a steady 7.5 × 10³ m/s. Calculate the time for one orbit.

Step 1. Start from v = 2πr/T. T is in the denominator and v is the lone term on the other side, so swap them in one step (cross-multiplication).

$$v = \frac{2\pi r}{T} \quad \Longrightarrow \quad T = \frac{2\pi r}{v}$$

One swap, not × T then ÷ v.

Units: $\frac{\text{m}}{\text{m s}^{-1}} = \text{s}$ ✓

Step 2. Substitute, with units.

$$T = \frac{2\pi \times 7.0 \times 10^{6}\ \text{m}}{7.5 \times 10^{3}\ \text{m s}^{-1}} = 5864\ \text{s}$$

Step 3. The data are to 2 s.f., so give 2 s.f. Divide by 60 s per minute as a sense check.

$$T = 5.9 \times 10^{3}\ \text{s} \approx 98\ \text{min}$$

Low orbits take about 1½ h: plausible.

Units: $\frac{\text{s}}{\text{s min}^{-1}} = \text{min}$ ✓

Answer: $T = 5.9 \times 10^{3}\ \text{s}$ (about 98 minutes)

05 Common mistakes

✗ **Wrong:** Constant speed means no acceleration.
✓ **Right:** The direction changes, so the velocity changes: it accelerates towards the centre.

✗ **Wrong:** A released ball flies outwards along the radius.
✓ **Right:** It moves off along the tangent at the point of release.

✗ **Wrong:** Use the diameter in v = 2πr/T, or T in minutes.
✓ **Right:** r is the radius in metres; T is the period in seconds.

✗ **Wrong:** A "centrifugal force" pushes the object outwards.
✓ **Right:** The resultant force points inwards. Feeling thrown out is your body carrying on in a straight line.

✗ **Wrong:** Draw centripetal force as an extra arrow beside tension or gravity.
✓ **Right:** It is the resultant of those real forces, not another force.

06 Quick recall

Q1 In which direction does the resultant force on an object in uniform circular motion act?

Q2 What provides the centripetal force that keeps the Moon in orbit?

Q3 Mass and radius stay the same but the force increases. What happens to the speed?

Q4 Mass and speed stay the same but the force increases. What happens to the radius?

Q5 Two cars take a bend at the same speed. One has twice the mass. Which needs more friction?

Q6 Calculate the speed of a point 0.30 m from the centre of a wheel that turns once every 0.50 s.

Answers 1. Towards the centre of the circle, at 90° to the velocity. · **2.** The gravitational attraction of the Earth. · **3.** The speed increases. · **4.** The radius decreases: the object moves in a tighter circle. · **5.** The heavier car: more mass needs more force for the same speed and radius. · **6.** v = 2π × 0.30 m ÷ 0.50 s = 3.8 m/s

07 If you remember nothing else

1. Steady speed in a circle still means acceleration: the direction of the velocity keeps changing.
2. That needs a resultant force towards the centre, at 90° to the velocity, supplied by tension, friction or gravity.
3. More force: faster at the same radius, or a tighter circle at the same speed. More mass: more force needed.
4. Remove the force and the object leaves along the tangent, not outwards.
5. Average orbital speed: v = 2πr/T, with r in metres and T in seconds.